Primus Tech: Building smart green solutions for urban infrastructure with Google Cloud

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Cloud migration i
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for customers an
global expansion

About Primus Tech

Primus Tech is a leader in intelligent management systems for buildings and infrastructure. The Singapore-based company offers a comprehensive range of services and solutions that include AI-enabled Integrated Building Management Systems, Smart Facility Management Systems, Internet of Things, Data Analytics, Extra-Low Voltage Systems, Network Infrastructure and Infocomm Technology solutions. PrimusTech's customer-centric focus, deep domain expertise, agility in building new applications, continuous innovation and service

based on BigQue
analytics.

Google Cloud results

- Supports green mission for building industry under Google Cloud net-zero commitment
- Enables agile, scalable solutions with Cloud Storage managed service for unlimited data storage
- Builds customer trust with glitch-free cloud

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delivery excellence have resulted in enduring customer relationships. By putting people first, the company ensures that its work delivers tangible outcomes in line with customer goals. Its mission is to be the pacesetter in the digital transformation of the buildings and infrastructure of the future.

Industries: Technology

Location: Singapore

(https://cloud.google.com/compute/), Cloud Storage (https://cloud.google.com/storage/)

Google Cloud (https://cloud.google.com/)

Compute Engine (https://cloud.google.com/compute/)

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Cloud Storage (https://cloud.google.com/storage/)

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migration in under three weeks
- Forges a global expansion strategy based on data-driven innovation with BigQuery

According to the Climate Group, buildings account for 40 percent of global greenhouse gas emissions (https://www.theclimategroup.org/built-environment#:~:text=Overview,set%20to%20double%20by%202050.) and if left unchecked their carbon output is projected to double by 2050. The non-profit group says "energy efficiency has a vital role to play in lowering energy demand, reducing emissions and driving the clean energy transition."

Singapore-based Primus Tech (https://primustech.com.sg/) is playing a proactive role in the global mission for a net-zero world through AI-enabled smart solutions for buildings and infrastructure that create operational efficiencies to help companies on the decarbonization journey. The firm offers a range of solutions that include integrated building management systems, smart facility management systems, Internet of Things (IoT), and data analytics.

To succeed, the company needs to process massive amounts of data from client building assets, and integrate them in its computing environment to automate the optimal functioning of all building operations. This encompasses elevators to ventilation and air conditioning.

Primus Tech began operations in an on-premise environment and as its business grew throughout Singapore and across Southeast Asia, it found itself coming up against the limitations of providing real-time optimizations with bare-metal servers. These needed constant hardware provisioning, and lacked flexibility to scale up and down. The firm also faced cost and logistical challenges storing the vast amounts of data flooding into its systems. The company decided to embark upon a cloud migration, for the seamless scalability and managed data storage that it would empower it to meet customer needs.

After considering major options, Primus Tech tapped Google Cloud (https://cloud.google.com/) thanks to its combination of agility and system resilience. It was also won over by the Google Cloud commitment to being a net zero cloud operation. This was a key consideration in the firm's mission to drive energy optimizations of customers through data-driven, real-time maintenance.

In particular, Primus Tech chose a combination of Compute Engine (https://cloud.google.com/compute) and Cloud Storage (https://cloud.google.com/storage/) as the foundation of its transformed digital infrastructure platform. Virtual machine (VM) automation with Compute Engine alongside unlimited data storage on Cloud Storage, for retrieval anytime, brought Primus Tech the scalable agility it needed to succeed. With proactive Google Cloud support, Primus Tech carried out a comprehensive Google Cloud migration in under three weeks with zero downtime.

The results surpassed Primus Tech's expectations, as the cloud architecture helped it drive significant customer operational efficiencies while yielding up to 14 percent energy saving on average for customers.

"Supporting a net-zero future through smart infrastructure systems is a core part of our mission," says Alan Teo, Assistant Vice President, Strategic Alliance Management, Primus Tech. "The frictionless scalability of Compute Engine and Cloud Storage has been transformative in driving energy savings compared to an on-prem environment. Google Cloud's net zero commitment itself helps us support a more sustainable future in partnership with clients."

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Unleashing the power of smart building solutions with scalable cloud connectivity

The enterprise value of Primus Tech revolves around Sirius+, its proprietary software system. It's a platform that offers cloud connectivity to access real-time data and key performance metrics for operational savings and value management of building assets. This enables customers to monitor and optimize their energy and carbon usage. Primus Tech also runs its Sirius Low Carbon City on Google Cloud, which measures the carbon emission of buildings and uses AI to reduce building energy consumption.

Sirius+ leverages AI algorithms to integrate all systems within a facility, including heating, lighting, alarms, and water usage, for a 360 degree view that enables energy optimization. A key use case for Primus Tech's solution is temperature regulation in Singapore buildings, which are often set to coolerthan-normal temperatures in the island state's hot and muggy weather.

Sensors enable the Sirius+ system to monitor weather conditions to provide highly granular automated adjustments of temperature controls to a fixed set point, according to ambient temperatures and forecasts, for considerable energy savings. The system maintains the temperature set point at between 23 and 26 degrees celsius by deploying sensors to gauge up to hundreds of indoor and outdoor factors, such as humidity and air circulation.

Handling the task with on-prem servers on the client site leads to drawbacks such as the need to constantly provision servers and operating cooling systems that increase carbon footprint. The need to manually scale up and down computing power in an on-premise environment leads to cost, labor and operational inefficiencies and ends up resulting in energy waste.

"By switching from on-premise servers to Compute Engine for scalable VMs for these data tasks, as well as Cloud Storage to ingest unlimited data for anytime retrieval, we are able to drive a carbon neutral mission with granular temperature controls in Singapore buildings," says Meng Khim Neo Chief Digital Officer, Primus Tech.

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Building the future of smart infrastructure solutions with

Google Cloud

Looking into the future, Primus Tech is planning to expand its solution around the world. It already serves several clients that have a global footprint. This is enabling it to collect troves of valuable data from points around the planet to store in Cloud Storage and crunch through its proprietary algorithms. It can thereby drive new data insight on energy consumption and carbon emissions on a worldwide level on Google Cloud.

Primus Tech foresees the adoption of Google Cloud machine learning tools such as BigQuery (https://cloud.google.com/bigquery) as an intelligent data warehouse, as it moves to expand in more international markets and collects more data. This will further open a new world of tech innovation opportunity.

"We foresee continued creative collaboration with the Google Cloud team, including exploration of BigQuery possibilities, to innovate the future of smart building and infrastructure systems," says Neo. "The sky's the limit as far as possibilities are for enabling a net-zero future in the urban building space."

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